Rajat Ghosh, Ph.D.

+1 (404) 697-5789 — rajat.ghosh11@gmail.com — linkedin.com/in/i-am-rajat

Professional Summary

Technical leader with 12 years of expertise in reinforcement learning, optimal control, Large Language Models (LLM), AI safety, and retrieval augmented generation (RAG). Demonstrated success in architecting scalable, impactful AI-driven applications and leading multidisciplinary teams.

Skills

Domains: Large Language Models, ML Architecture, Reinforcement Learning, Retrieval Augmented Generation, API Design, LLMOps, Optimal Control
ML Libraries: HuggingFace, Transformers, Gym, PyTorch, Jax
Databases: SQL, Milvus, Elasticsearch
Languages: C++, Python, Go, JavaScript, Bash, Java
Technologies: PySpark, Docker, Prometheus, Jenkins, Postman
Platforms: AWS, GCP, NVIDIA

Professional Experience

Staff Data Scientist — Nutanix, May 2021 – Present

- Architected internal coding assistant agents utilizing open-source LLMs, leading a team of 3 engineers to enhance productivity by 20% ($100M annual impact).
- Developed retrieval augmented generation (RAG)-based support assistants using fine-tuned LLMs with instruction tuning and RLHF, integrated with open-source VectorDB ($20M annual impact).
- Engineered reinforcement learning strategies for distributed infrastructure management including Kubernetes, Cassandra, storage tiering, data warehousing, load balancing, and JVM optimization.
- Created Shapley-inspired explainability metrics for RL recommendation systems.
- Deployed object detection inference APIs using YOLO v5 for retail applications.

Co-Founder & CEO — AdeptDC, Inc., July 2015 – April 2021

- Founded startup leveraging Ph.D. research to optimize data center operations through real-time monitoring and reinforcement learning-based dynamic control.
- Managed a 7-member team, successfully generating proof-of-concept revenue with Fortune 500 companies.
- Secured ~$1M funding from venture capital and NSF grants; successfully exited via acquisition by a major cloud company.

Education

- **Ph.D.**, Georgia Institute of Technology, Atlanta, 2013
- **M.S.**, Georgia Institute of Technology, Atlanta, 2010

- **B.S.**, Indian Institute of Technology, Kharagpur, 2008

Honors and Awards

- NSF Small Business Innovation Research Award, 2015
  - Real-time software-based cooling optimization for data centers.
- Georgia Research Alliance Phase IA Award, 2014
- NSF I-Corps Award, 2014

Publications

[1] A. Singhal, R. Ghosh, R. Mundra, H. Dadlani, and D. Dutta, "Code2json: Can a zero-shot llm extract code features for code rag?" In *ICLR 2025 Third Workshop on Deep Learning for Code*.

[2] S. Ghosh, H. Frase, A. Williams, *et al.*, "Ailuminate: Introducing v1. 0 of the ai risk and reliability benchmark from mlcommons," *arXiv preprint arXiv:2503.05731*, 2025.

[3] V. Bhargava, R. Ghosh, and D. Dutta, "Cpp-ut-bench: Can llms write complex unit tests in c++?" *arXiv preprint arXiv:2412.02735, Neurips Statistics in LLMs Workshop*, 2024.

[4] A. Khera, R. Ghosh, and D. Dutta, "Efficient alignment of large language models via data sampling," *arXiv preprint arXiv:2411.10545, NeurIPS ENLSP. PMLR*, 2024.

[5] B. Vidgen, A. Agrawal, A. M. Ahmed, *et al.*, "Introducing v0. 5 of the ai safety benchmark from mlcommons," *arXiv preprint arXiv:2404.12241*, 2024.

[6] R. Ghosh and Y. Joshi, "Proper orthogonal decomposition-based modeling framework for improving spatial resolution of measured temperature data," *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 4, no. 5, pp. 848–858, 2014.

[7] R. Ghosh and Y. Joshi, "Rapid temperature predictions in data centers using multi-parameter proper orthogonal decomposition," *Numerical Heat Transfer, Part A: Applications*, vol. 66, no. 1, pp. 41–63, 2014.

[8] R. Ghosh and Y. Joshi, "Error estimation in pod-based dynamic reduced-order thermal modeling of data centers," *International Journal of Heat and Mass Transfer*, vol. 57, no. 2, pp. 698–707, 2013.

[9] R. Ghosh, G. A. Buxton, O. B. Usta, A. C. Balazs, and A. Alexeev, "Designing oscillating cilia that capture or release microscopic particles," *Langmuir*, vol. 26, no. 4, pp. 2963–2968, 2010.

Patents

Lead Inventor, U.S. Utility Patent (2019): Systems and Methods for Intelligent Controls in Data Centers, reducing operational costs by 35%.

Invited Talks

"Agentic AI: Future Challenges and Opportunities," AAAI Workshop, March 2025

Teaching Experience

- Thermodynamics (ME 3332), Georgia Tech, 2013

- Experimental Methodology (ME 3057), Georgia Tech, 2012-2013
- Introduction to Energy Systems Engineering (ME 4803), Georgia Tech, 2012
- Creative Decisions and Design (ME 2110), Georgia Tech, 2009

Review & Service

Reviewer for NeurIPS, AAAI, IEEE CPMT Journal, InterPACK, Semi-Therm, I-Therm.

Press Coverage

Featured in TechRepublic, CIOReview, IT Business Edge for innovations in data center efficiency.